

Chapter 8

Cameras

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Learning Outcomes

After studying this chapter, you are expected to be able to

1. Recognise some of the different types of cameras used in the real world, and the need for a digital camera in rendering
2. Describe the essential components of a camera including Position, Front, Up, Right, Pitch, Yaw and Roll.
3. Explain how a Model-Viewer Camera works including the elements that the user can interact with
4. Calculate the Model-Viewer Camera Position using Pitch and Yaw, distance, and the unit sphere
5. Illustrate how a Fly-Through Camera works including the elements that the user can interact with
6. Compute the Right and Up camera vectors using Cross Product
7. Calculate the Fly-Through Camera Position using vector addition and subtraction with the Front and Right vectors

Introduction

In this chapter, we will learn about cameras.

We have already looked at how to transform the vertices to a camera position using the view matrix in chapter 6. We can then apply one of the basic transformations to the view matrix to move and rotate the camera. This will work, but what we really need is more intuitive way to control a camera.

There are many different types of cameras which are common in film and television, including

1. a camera attached to a plane to get action shots of planes flying and maneuvering
2. a camera attached to a helmet so that the viewpoint is the same as a person's eyes
3. a camera that attaches to a moving crane and used for filming Formula 1
4. a camera that records a football match from above the pitch at the top of the stadium and follows the action

to name only a handful of the common cameras in use today.

As you can see, the idea of an intuitive and useful camera isn't really a science but rather an art form.

It is also specific to the problem of what needs to be photographed, whilst keeping in mind the needs of the viewer.

In COMP3069, we will look at 2 types of cameras, which can be called

1. A Model-Viewer Camera
2. A Fly-Through Camera

Once we have implemented one of these cameras, we can use it to intuitively place the camera in the scene to render the geometry.

We can also use it as a way of interacting with graphics and moving the camera interactively.

More importantly, we will be able to use the camera to view very specific points in the scene from specific angles, intuitively, which will be very useful for testing and debugging graphics programming.

The Essentials of a camera

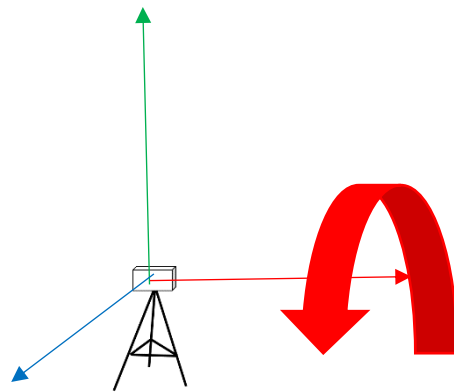
For our purposes, a camera involves

1. A Position in 3D space
2. A Forward Direction, a 3D vector
3. A Right Direction, a 3D vector
4. An Up Direction, a 3D vector

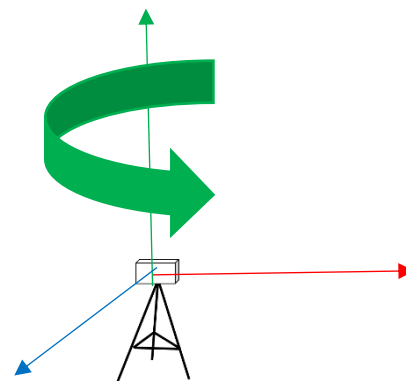
A camera position is its location in world space.

Calculating a camera direction can be achieved using Euler angles, Pitch, Yaw and Roll.

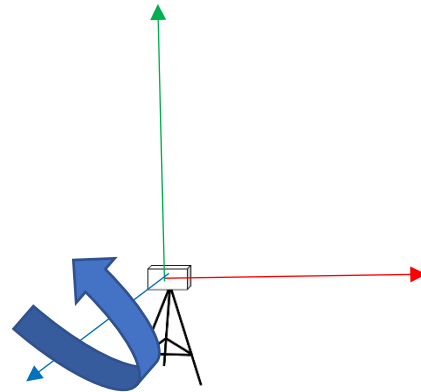
Pitch rotates the camera on the x-axis. Think of pointing an airplane's nose at the ground or at the sky, but keeping the wings parallel with the horizon.



Yaw rotates the camera on the y-axis. Think of pointing an airplane's nose from left to right along the horizon, whilst keeping the wings parallel with the ground.



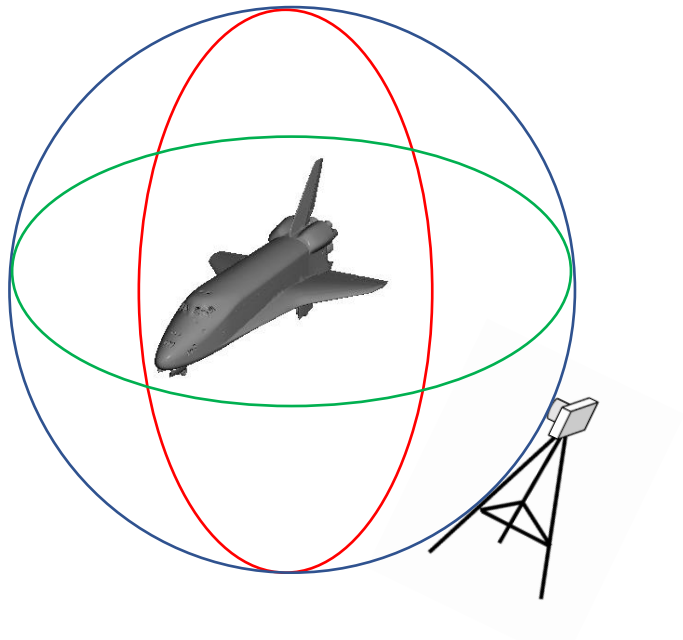
Roll rotates the camera on the z-axis. Think of tilting an airplane's wings left and right, but not changing the direction the nose is pointing.



The camera's Position, Pitch, Yaw and Roll, and Front, Right and Up, can be calculated or combined in different ways. This is the subject of this week's lecture.

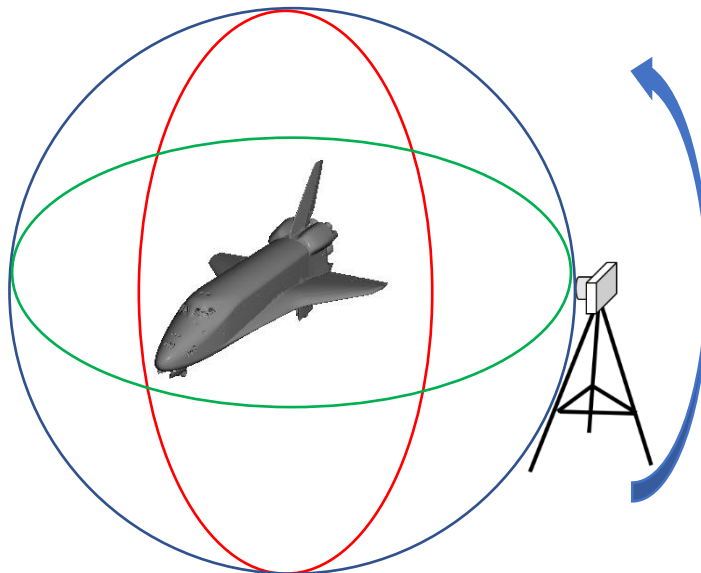
Model-Viewer Camera

The first type of cameras that we will look at is called a Model-Viewer camera.

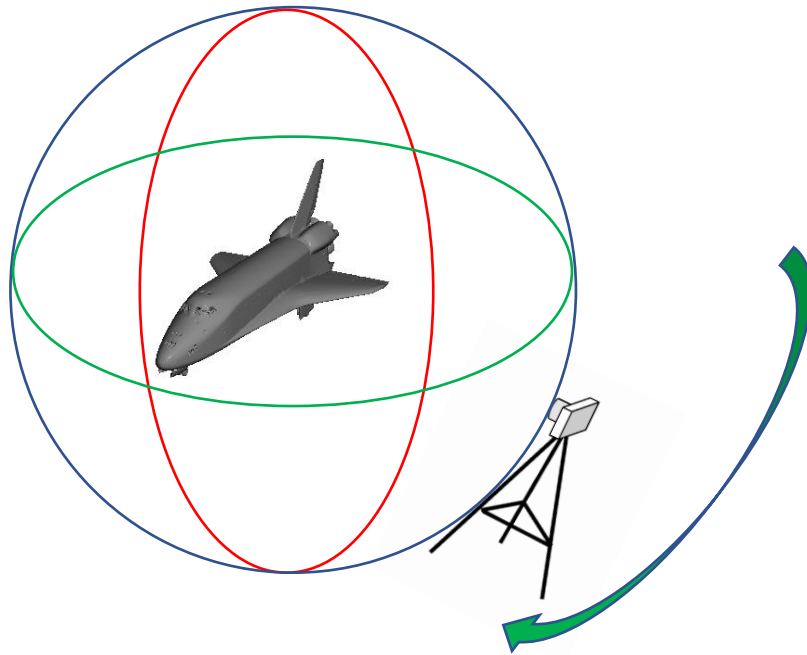


As can be seen in the image, the camera constantly looks at the same target which is the model being viewed. We leave the model at the origin (0, 0, 0).

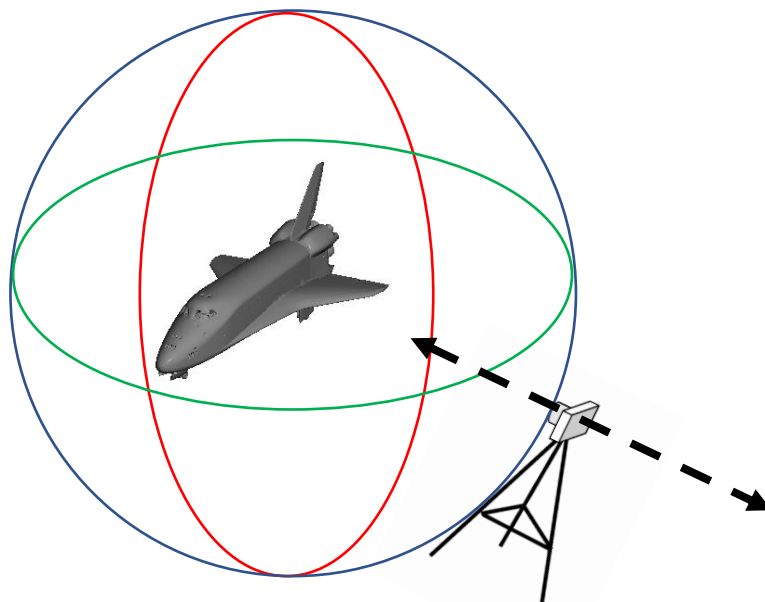
The user can modify the camera Pitch, which will move the camera up and down along the latitude, around an imaginary sphere surrounding the model.



The user can modify the camera Yaw, which will move the camera left and right along the longitude, around an imaginary sphere surrounding the model.



The user can also modify the distance of the camera from the model forward and backward, which changes the radius of the imaginary sphere.



In this way the camera can be manipulated to view the model from all angles and distances.

The Model-Viewer camera is implemented using a unit sphere.

We can extend the algorithm for calculating points on a unit circle in 2-Dimensions, so that it calculates points on a unit sphere in 3-Dimensions.

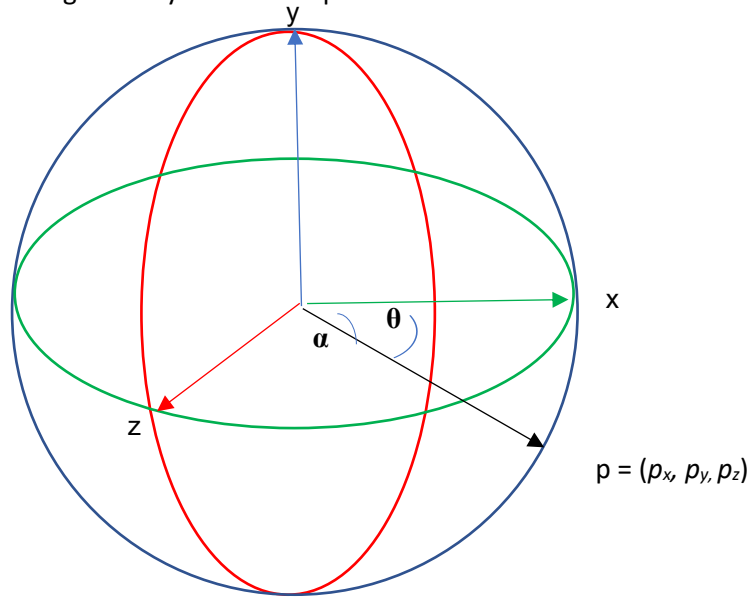
If $\mathbf{p} = (p_x, p_y, p_z)$ is a unit vector in 3D, the trigonometric functions are

$$p_x = \cos \theta * \cos \alpha$$

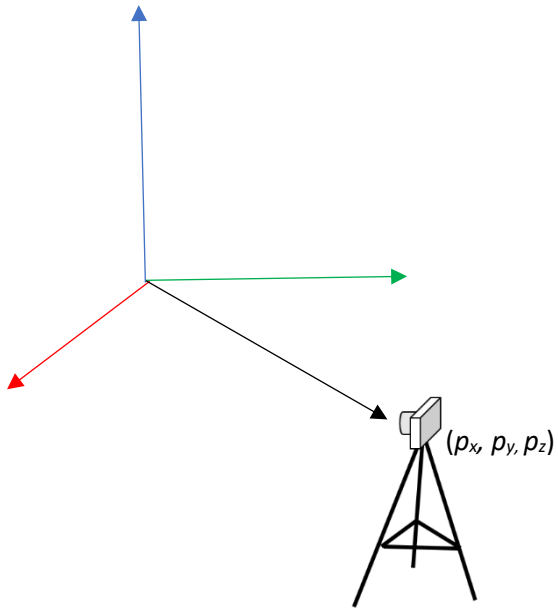
$$p_y = \sin \alpha$$

$$p_z = \sin \theta * \cos \alpha$$

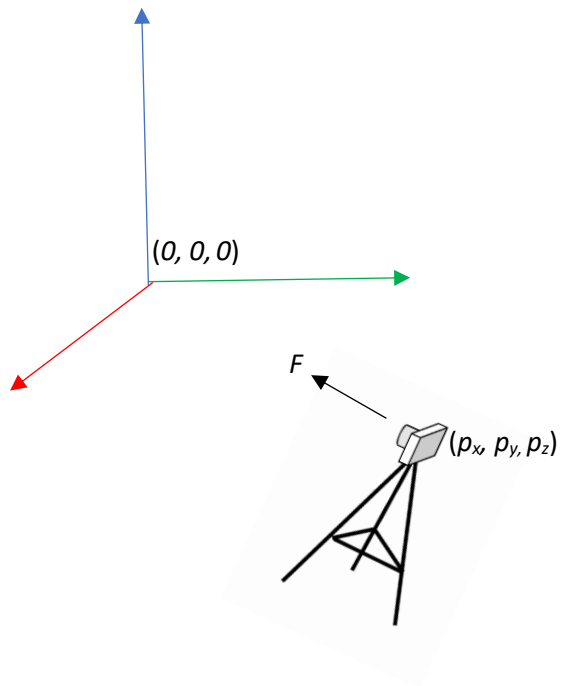
The geometry is the unit sphere



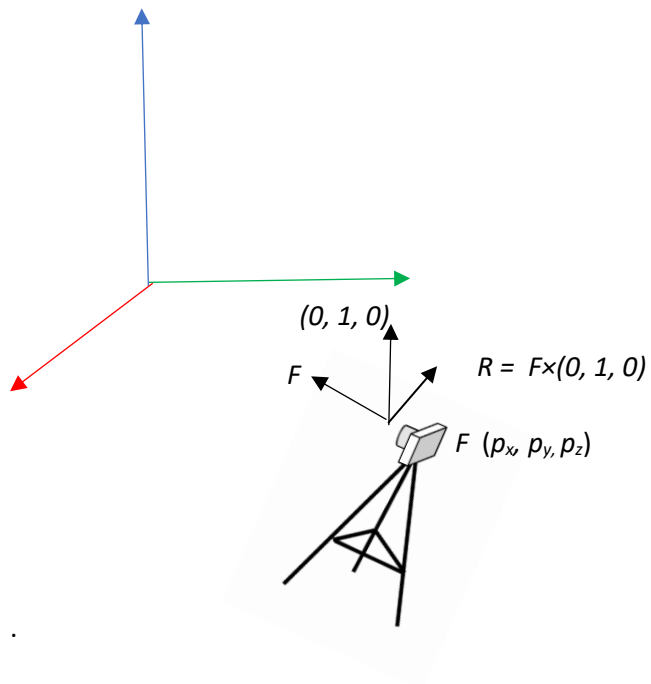
If we use $\theta = \textit{Yaw}$ and $\alpha = \textit{Pitch}$, then the camera Position is the calculated point on the unit sphere.



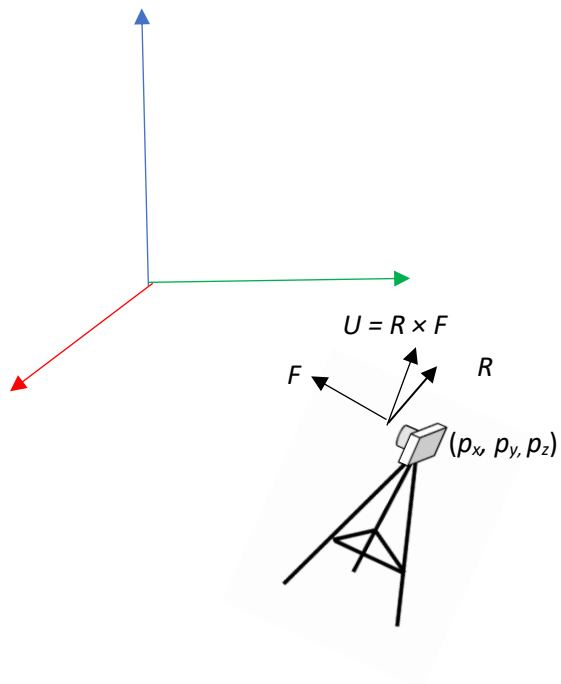
The Front direction of the camera is a vector pointing at the origin of the space.



The Right direction of the camera is the vector which is the result of cross product of camera Front with absolute up direction $(0, 1, 0)$.



The Up direction of the camera is the vector which is the result of cross product of camera Right with camera Forwards.



In this way, we can calculate the camera Position, Front, Right and Up components using the camera Yaw and Pitch.

Fly-Through Camera

The second type of cameras that we will look at is called a Fly-Through camera.

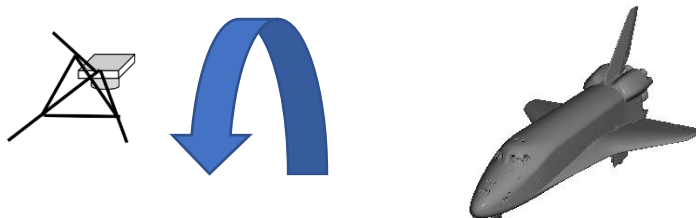
As can be seen in the image, the camera has a Position in world space and a focus direction Front, which may or may not be the model which is being viewed.



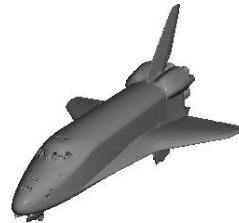
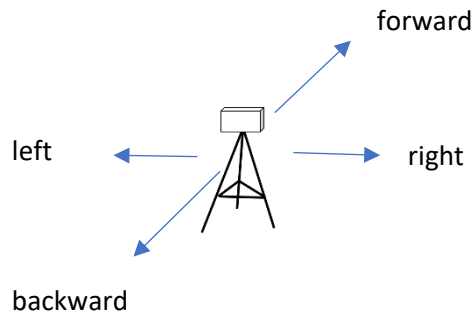
The user can modify the Yaw to alter the focus direction



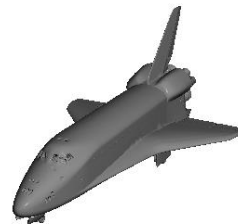
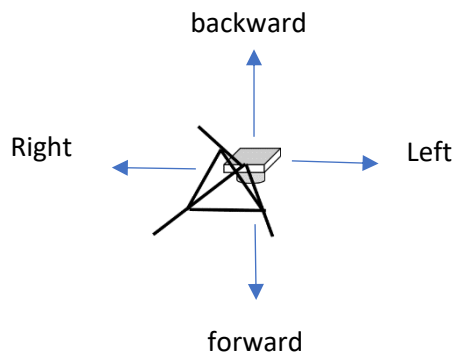
The user can also modify the Pitch to alter the focus direction



The user can move the camera forwards, backwards, left and right.

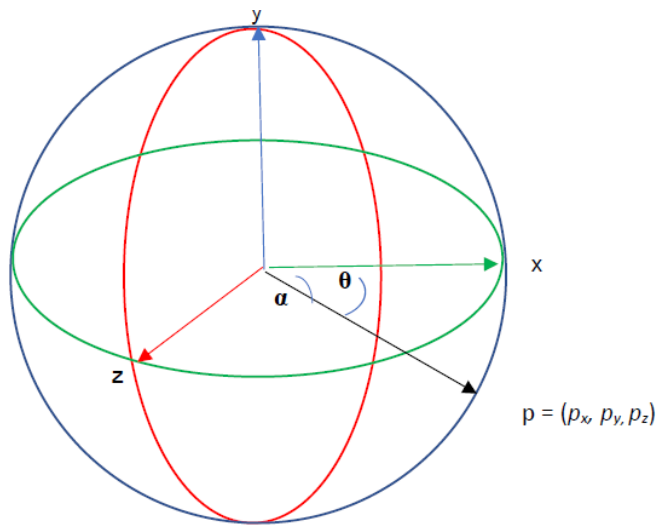


And the directions forwards, backwards, left and right are relative to the focus direction Front.



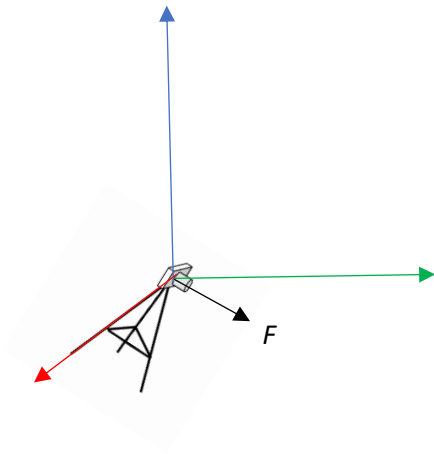
In this way, the camera can be manipulated to fly through the scene in all directions and can look in any direction.

The Fly-Through camera is also implemented using the unit sphere

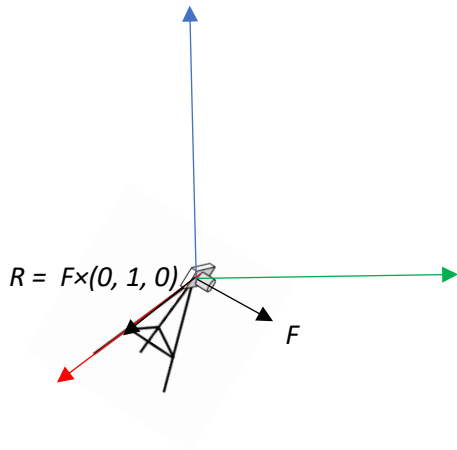


For the Fly-Through Camera, we need to imagine that the centre of the unit sphere is at the location of the camera.

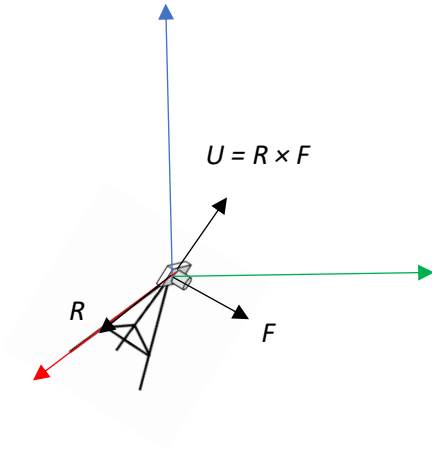
The focus direction Front of the camera is the vector which is the point on the unit sphere. Since it is a unit sphere, then this point is also a unit direction vector.



The Right direction of the camera is the vector which is the result of cross product of camera front with absolute up (0, 1, 0).



The Up direction of the camera is the vector which is the result of cross product of camera Right with camera Front.

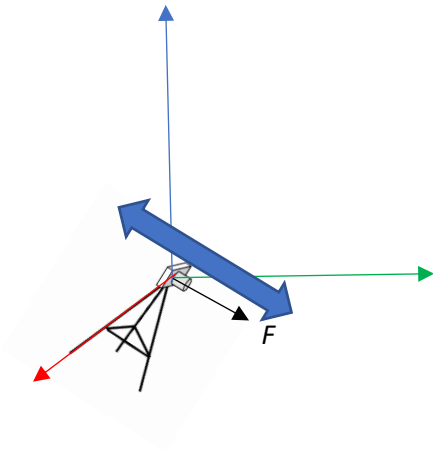


The position of the camera can then move forwards by adding camera front

$$Position = Position + Front$$

or backwards by subtracting camera front.

$$Position = Position - Front$$

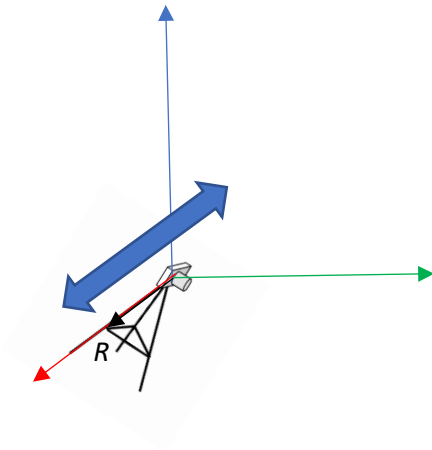


The position of the camera can then move right by adding camera right

$$Position = Position + Right$$

or left by subtracting camera right.

$$Position = Position - Right$$



In this way, we can calculate the camera Position, Front, Right and Up components using the camera Yaw and Pitch.